Answer Key - Astronomy - Graduate Level Assessment 14 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** B
   Options: A) names for the layers of gaseous and metallic hydrogen deep within the planet.; B) alternating bands of rising and falling air at different latitudes.; C) alternating regions of charged particles in Jupiter's magnetic field.; D) regions of the plasma torus created by ions from Io's volcanoes
   Note: The belts and zones of Jupiter are characterized by alternating bands of rising and falling air due to the planet's rapid rotation and convection processes, which create distinct atmospheric circulation patterns at different latitudes.

2. **Answer:** C
   Options: A) Brighter; B) Dimmer; C) Higher in the sky; D) Lower in the sky
   Note: As you travel northward from the United States into Canada, you are moving closer to the North Star (Polaris), which is located nearly directly above the North Pole, causing it to appear higher in the sky.

3. **Answer:** C
   Options: A) You will not be able to see these two stars at all.; B) You will see two distinct stars.; C) The two stars will look like a single point of light.; D) The two stars will appear to be touching looking rather like a small dumbbell.
   Note: When the angular separation of two stars is smaller than the angular resolution of the human eye, the eye cannot distinguish the individual stars, causing them to blend together and appear as a single point of light.

4. **Answer:** D
   Options: A) in the center of the asteroid belt; B) on orbits that cross Earth's orbit; C) surrounding Jupiter; D) along Jupiter's orbit $60^\circ aheadofandbehindJupiter$
   $Note:TheTrojanasteroidsarelocatedatstableLagrangepointsinJupiter'sorbit, specifically60degrees$

4. **Answer:** A
   Options: A) asteroids would orbit with a period half that of Jupiter; B) asteroids would orbit with a period twice that of Jupiter; C) asteroids would orbit with a period twice that of Mars; D) A and B
   Note: Kirkwood gaps occur in the main asteroid belt at locations where the gravitational influence of Jupiter causes resonances that destabilize the orbits of nearby asteroids, specifically at positions where their orbital period is a simple fraction, such as half, of Jupiter's orbital period.

True / False

6. **Answer:** False
   Options: A) True; B) False
   Note: Mercury's atmosphere is extremely thin and primarily lost due to solar wind and its lack of significant gravity, rather than volcanic heating, which does not play a significant role in replenishing it.

7. **Answer:** True
   Options: A) True; B) False
   Note: Organisms living in extreme environments such as deep-sea volcanic vents and hot springs share characteristics with early life forms, which likely evolved in similar conditions, suggesting they resemble the common ancestor of all life more closely than other organisms.

8. **Answer:** True
   Options: A) True; B) False
   Note: Moons, through their gravitational interactions, can stabilize the orbits of particles within planetary rings, create gaps by exerting gravitational influence that clears certain areas, and shape the edges of the rings by pulling on the material at these boundaries.

9. **Answer:** False
   Options: A) True; B) False
   Note: Jupiter's magnetic field is actually about 20,000 times stronger than Earth's magnetic field at its surface, but the statement is misleading as it does not specify the context or location where this comparison is made, leading to potential misconceptions about the differences in magnetic field strength.

10. **Answer:** False
    Options: A) True; B) False
    Note: The statement is false because a solar day is determined by the time it takes for a planet to rotate on its axis relative to the sun, and a prograde rotation typically results in shorter solar days, while orbital synchronization would imply a different relationship between the planet's rotation and its orbit around its star.

Long Form Response

11. **Answer:** The acceleration of a truck on Mars compared to Earth, assuming no friction, would be the same due to the fundamental principles of Newton's second law of motion, which states that acceleration is independent of the gravitational force acting on an object when only considering the net force applied. On both Mars and Earth, if the same force is exerted on the truck, it will accelerate at the same rate, regardless of the differing gravitational forces. This conclusion highlights that while gravitational force affects the weight of the truck and the normal force acting on it, it does not influence the acceleration when friction is absent. Thus, the key takeaway is that in

a frictionless environment, the acceleration of an object is determined solely by the net external force acting upon it, reaffirming the universality of Newton's laws across different gravitational contexts.
Note: correct because Newton's second law of motion asserts that acceleration is determined solely by the net force acting on an object and its mass, and since this relationship does not depend on gravitational force in the absence of friction, a truck would accelerate at the same rate on Mars as it would on Earth when the same force is applied.

12. **Answer:** In astronomy, a black body is defined as an idealized physical object that absorbs all electromagnetic radiation incident upon it, reflecting no light. This concept is significant in calculating stellar radiation as it serves as a benchmark for understanding the emission spectra of stars, allowing astronomers to apply Planck's law to determine the temperature and energy output of celestial bodies. The properties of black bodies facilitate the study of stellar evolution and the classification of stars, as their radiation characteristics provide insight into surface temperatures and luminosities. Consequently, understanding black bodies is crucial for interpreting observational data, enabling astronomers to draw conclusions about stellar composition and the underlying physical processes governing stellar behavior.
Note: correct because it accurately defines a black body as an idealized object that absorbs all radiation, emphasizes its crucial role in applying Planck's law for calculating stellar radiation and temperature, and highlights its importance in understanding stellar evolution and classification.

End of Answer Key